COM-480 Data Visualization

Milestone 2

Authors

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1 Project Goal

The goal of our project is to visualize diversity across the Museum of Modern Art (MoMA) collection and to help users discover underrepresented artists, works, and mediums in the collection. Specifically, we will focus on how representation in the collection varies according to gender and nationality and how this composition of the collection has evolved over time. Furthermore, we aim to produce an interactive and accessible platform, such that a general audience will be able to explore these characteristics without prior art knowledge.

2 Narrative Structure

Our interface is organized as a scroll-driven narrative. We guide the reader through a sequence of scenes in which each scroll step reveals one new layer of a central visualization and answers one specific question. The story moves from the shape of the collection as a whole, through geography and gender, to an open explorer, and finally to a personal discovery. A global decade range slider threads through all scenes, allowing the reader to filter every view simultaneously by time period.

2.1 Scene 1: The Collection Takes Shape

The first central visual is a grid of artistic mediums, with one square for each in the collection. Hovering over any zone reveals the exact count and percentage. Scroll steps progressively fill in each medium, beginning with Drawings & Prints (over 50%), followed by Photography and Architecture & Design, before revealing that Painting & Sculpture (what most visitors associate with MoMA) accounts for around 3% of works. The purpose of this scene is to correct the reader's prior assumptions about the collection before asking who made it.

2.2 Scene 2: Where Are These Artists From?

The second scene centers on an orthographic globe that begins dark and lights up progressively as the reader scrolls. The first step reveals that artists in the collection come from 129 countries. The second step ignites three bright spots, the United States, France, and Germany, and narrates that three countries account for over two thirds of all works. The rest of the world visibly dims in contrast. A decade scrubber then appears, inviting the reader to drag through seventy years of acquisitions; the key finding is that this geographic concentration has barely loosened over time. Clicking a highlighted country at any point reveals a randomly selected artist from that region in a side panel.

2.3 Scene 3: What About Gender?

The third scene opens with a line chart tracing the female share of credited works decade by decade. Early spikes in the 1910s and 1930s hint at brief openings that did not hold. For most of the mid-twentieth century the line stays low. A sustained rise begins in the 1980s and accelerates through the 1990s, before approaching

parity in the most recent acquisitions. A final scroll step splits the view by curatorial department, revealing that this progress has been deeply uneven; Media and Performance reached significant female representation as early as the 1960s, while Architecture & Design and Drawings & Prints remained near zero until the 1980s and beyond.

2.4 Scene 4: Does Your Medium Matter?

The fourth scene transitions from guided narration to open exploration.

A small multiples grid, one panel per medium, shows the gender and nationality breakdown within each medium for the selected period. The grid is fully filterable by medium, department, and decade, and linked to the globe selection from Scene 2. This section serves as the open explorer: the structural patterns established in the previous scenes become personally navigable rather than passively received.

2.5 Scene 5: Who Are You In The Collection?

The final scene inverts the direction of the piece. A short prompt frames the transition from aggregate patterns back to individual people.

The reader is presented with a three-question quiz (preferred decade, world region, and medium) and matched to an underrepresented artist whose attributes overlap with their selections. The artist card displays name, nationality, gender, one artwork title and date, and the anchoring statistic showing how few works that artist has in the collection compared to the most-represented figure. The scene ends on a human face rather than a chart, which is where the emotional weight of the narrative lands.

3 Tools & Lectures

Our project primarily utilizes the Svelte and D3.js frameworks on top of the native HTML, CSS, and JavaScript. The lectures we will refer to are as follows.

1. Data - Used to process and clean the data from the MoMA dataset
2. D3.js - Integrated within Svelte components to build the basic skeleton of our visualization
3. Interactions - Used as a reference to design our visualization's user experience
4. Perception Colors - Used to design a cohesive and easily digestible color palette across the project.

4 Additional Ideas

Time permitting, we will add a section to the end of the visualization shedding light on museum visitor insights into the diversity representation of the collection. To do so, we will scrape Google Reviews on the Museum of Modern Art, train an LLM to extract sentiment scores regarding diversity representation, and provide a visualization of overall sentiments.

As well, a possible challenging aspect of our visualization is animating each visualization based on scroll, such that scrolling through the site automatically triggers focus on different elements of each visualization that appears. If time is too tight, this feature can be omitted without sacrificing the meaning of this project.

Finally, time permitting, we would like to implement Scene 4 using a stacked pictograph of the 5 most prevalent mediums as illustrated below, instead of the small multiples visualization.

Figure 1: High fidelity illustration of alternative Scene 4 implementation.

5 Prototype

A functional prototype with the globe, gender charts, nationality streamgraph, department chart, and decade slider is currently deployed at:

`https://com-480-data-visualization.github.io/DArts/`

6 Sketches

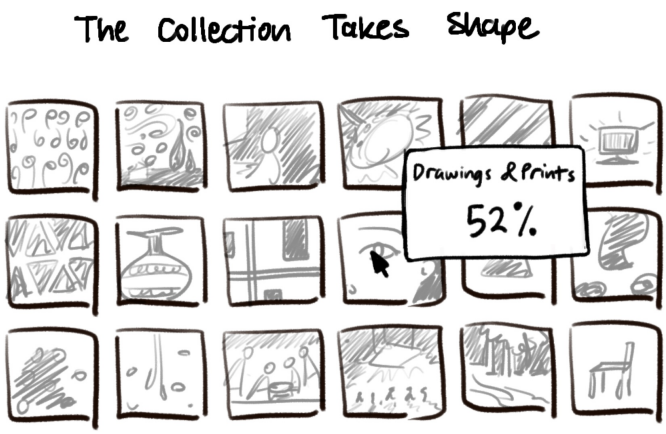

Figure 2: Scene 1 Sketch

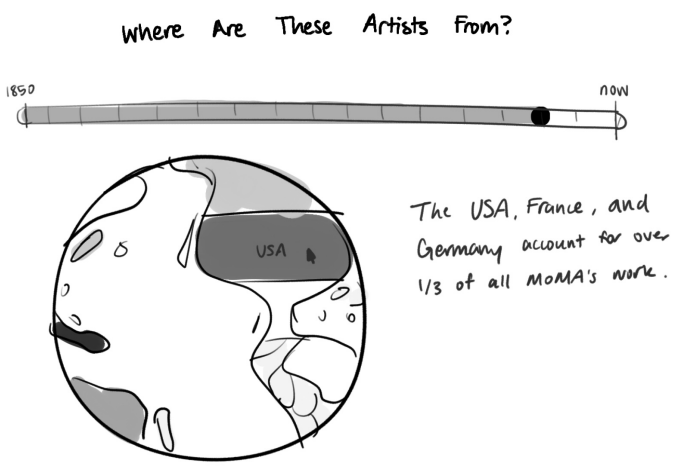

Figure 3: Scene 2 Sketch

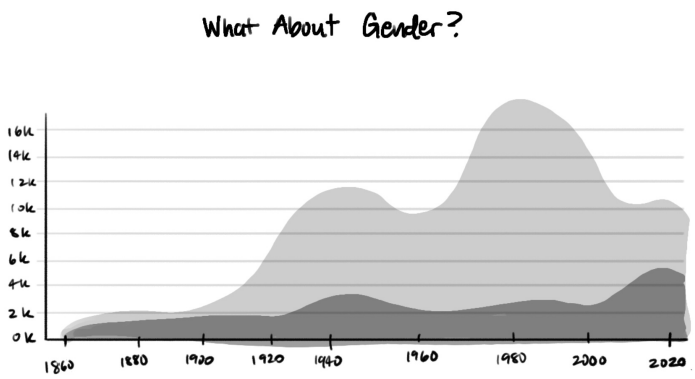

Figure 4: Scene 3 Sketch

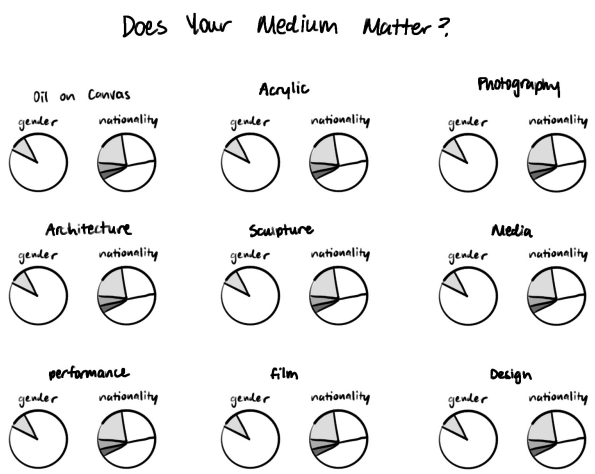

Figure 5: Scene 4 Sketch

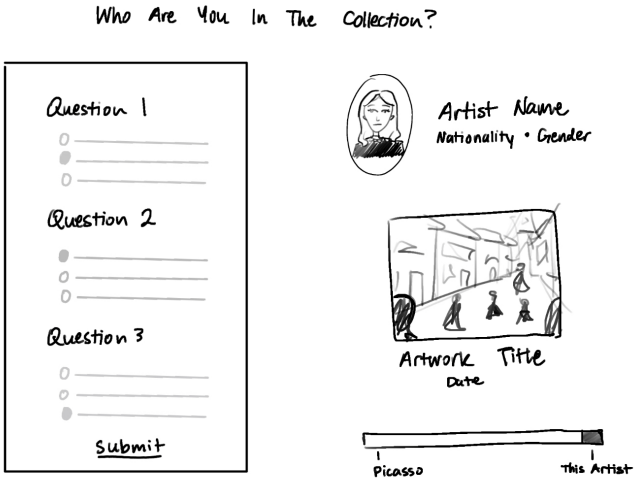

Figure 6: Scene 5 Sketch